#7 Annotated Bibliography

Questions	☑ ~~SDG's~~ ☑ ~~What are the real climate implications of constantly throwing away material?~~ ☑ ~~What are different ways established STEM programs currently uplift underprivileged areas? How can our app add to that?~~
Roadmap	☐ App viability ☑ ~~Climate implications~~ ☑ ~~Implications on underprivileged areas~~ ☐ Formulating user questions
Helpful Keywords	"STEM scale of waste" + "climate implications" "resource disparity in STEM education" "STEM lack of redistribution of materials"

Source #1

Source Name, Author, Citation Link	Source Name: **Why STEM Education Fails in Low-Income Schools and How to Fix It** Author: Sean Newman Maroni Citation Link: https://www.betabox.com/news/why-stem-education-fails-in-low-income-schools
Content Summary	1. **Resource Gap** A resource gap fuels the difference in education/opportunities for underprivileged schools. A huge reason being the lack of quality materials.
Connection to App	• Shows how the lack of equitable STEM materials affects opportunities and education in STEM • SDG 4, 10
Questions	1. **Would we be able to control this surplus and allocate materials to people who need it? (and how)**

Source #2

Source Name, Author, Citation Link	Source Name: **INTERSECTIONALITY IN EDUCATION: ADDRESSING THE UNIQUE CHALLENGES FACED BY GIRLS OF COLOUR IN STEM PATHWAYS** Author: Ndubuisi Sharon Amaka Citation Link: https://www.doi.org/10.56726/IRJMETS64288
Content Summary	1. **Resource Gaps Fuel Harmful STEM Stereotypes on Women** For instance, a study by the National Science Foundation found that students in well-funded schools are three times more likely to participate in lab-based experiments than their peers in low-income schools. 2. **The Impact of a Solution** By transforming the educational landscape, this research highlights the potential to not only improve outcomes for girls of colour but also to foster broader equity and diversity within STEM fields.
Connection to App	• Resource gaps aren't just denying opportunities, it can perpetuate harmful stereotypes on women, especially those of color • Having an app that could provide resources to underserved areas could help foster equity and diversity in STEM fields
Questions	

Source #3

Source Name, Author, Citation Link	Source Name: **Making the Grade 2024: Education Funding Disparities Persist as Some States Prioritize Tax Cuts and Privatization** Source Author: Education Law Center Citation Link: https://edlawcenter.org/making-the-grade-2024-education-funding-disparities-persist-as-some-states-prioritize-tax-cuts-and-privatization/
Content Summary	1. **Funding Disparity** **"**Average funding disparities between the highest and lowest funded states were persistently between $13,000 and $14,000 per pupil"
Connection to App	• HUGE discrepancies within state to state • Shows the need for schools with the privilege of having excess or out of date material to give back to the community
Questions	1. **Does our app help soften the blow at all?**

Source #4

Source Name, Author, Citation Link	Source Name: **Electronics Basic Information, Research, and Initiatives** Author: EPA (United States Environmental Protection Agency) Citation Link: https://www.epa.gov/electronics-batteries-management/electronics-basic-information-research-and-initiatives **CHECK OUT: EPA ADEPT Technology
Content Summary	1. **E-waste Awareness is VITAL** "A first step in using electronics sustainably involves educating consumers about sustainable purchasing choices. Learn about innovative approaches for product procurement and use." 2. **Reusing/Refurbishing Electronics STOPS e-Waste** "Preventing waste in the first place is preferable to any waste management option, including recycling. Donating used (but still operating) electronics for reuse extends the lives of valuable products and keeps them out of the waste stream for a longer period of time." 3. **Proven Benefits** "Donating electronics allows schools, nonprofit organizations and lower-income families to obtain equipment that they otherwise could not afford."
Connection to App	• Awareness component to the app ○ Suggestions when shopping for other electronics/components if other people are selling ○ Just facts in the app maybe? • SDG 12, 13
Questions	1. **What can awareness look like?**

STEM Uplift Programs/Donation Services

Program Name	Description	Pros/Cons Impact
Good360	Provides resources from corporate and individual donations to fuel STEM nonprofits	**Pros:** • Connects kids that can't afford the tech/resources with the goods **Cons:** • Only reaches the bandwidth of whatever the nonprofit works under • Not a total guarantee that resources could be allocated to your area
Kid Spark	Provides kits/learning materials to teach STEM to underserved areas	**Pros:** • Enables STEM accessibility in underserved areas • Provides free/reduced cost Kid Spark programs to over 20,000 kids across the U.S. **Cons:** • Not entirely free (grants/resources are not guaranteed and application based) • Only for U.S. Title 1 Schools
STEMChests	For every $5 donation or STEM kit purchased, another kit is donated to an underprivileged school in the greater D.C. area	**Pros:** • Donates STEM kits **Cons:** • Doesn't eliminate e-Waste • Relies on purchases from other people • Only works for the greater D.C. area
AZStRUT E-RECYCLEE	Refurbishes donated electronics and donates them students, schools, and nonprofits throughout Arizona	**Pros:** • Makes electronics accessible WHILE eliminating e-Waste **Cons:** • No direct

		conversation between the giver and the needer ● Limited to specific states ● Not really specialized electronics more over just standard technology